MIL-D-13684D (AR)
14 February 1969
SUPERSEDING
MIL-D-13684C (MU)
12 December 1963 and
MIL-D-46543A (MU)
12 December 1963

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, M48
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing of one type of electric initiating device designated as Detonator, Electric, M48.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-148 - Aluminum Foil
MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-650 - Explosives: Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition

FSC: 1375

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

7549133 - Detonator, Electric, M48 Assembly
8796522 - Marking Diagram and Sealing for Wood Packing Boxes

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 7549133 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, 25, D. C. 20402. Orders shall specify 49 CFR 171-179 (latest revision).)

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with applicable drawings and specifications.

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3.2 Detonator.- The detonator shall comply with all requirements specified on Drawing (dwg.) 7549133 and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Lead azide.- The moisture content of the lead azide at the loading station at the time of loading the detonator shall not exceed 0.50 percent when tested as specified in 4.4.1.1.

3.3.2 Pentaerythrite tetranitrate (PETN).- The moisture content of the PETN at the loading station at the time of loading the detonator shall not exceed 0.50 percent when tested as specified in 4.4.1.2.

3.4 Functioning.- The detonator assembly shall comply with the following requirements:

3.4.1 Time.- The detonator assembly shall fire on the first attempt within ten (10) microseconds when tested as specified in 4.4.2.1.

3.4.2 Output.- The detonator assembly shall fire on the first attempt and shall make a 5/8 inch minimum (min.) diameter perforation in the lead disc when tested as specified in 4.4.2.2.

3.4.3 Detonator assemblies that fail to fire on first attempt (Duds).- The detonator assembly shall not dud when tested as specified in 4.4.2.3.

3.5 Waterproofness.- The detonator assembly shall comply with the requirements of 3.4.2 after water immersion as specified in 4.4.2.2.

3.6 Electrical resistance between shorted lead wires and cup.- The electrical resistance between the shorted lead wires and the cup shall be not less than fifty (50) megohms when tested as specified in 4.4.3.

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3.7 Pull test of lead wires.- The lead wires of the plug shall withstand a tensile force of fifteen (15) pounds, minimum, applied axially between the plug and the lead wire, without indication of slippage of the lead wire in the plug when tested as specified in 4.4.5.

3.8 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.9 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

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b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that statements of findings on the lead azide and PETN and certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five (25) plug assemblies and twenty-five (25) complete sets of parts (i.e., 25 each of every component and every sub-assembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance for materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

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4.2.2 Inspections to be performed.- Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded detonator assemblies shall contain:

a. Parts of one lot interfix number from one supplier.
b. PETN from not more than one lot.

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c. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup (see dwg. 7549138 covering a detail of dwg. 7549133).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Thickness through bottom, min...	Gage	01001
102.	Outside diameter, maximum,(max.)	Gage	01002
103.	Dent or distortion.....	Visual	01003
Minor:	AQL 1.00 percent		
201.	Inside diameter, min.....	Gage	01004
202.	Evidence of poor workmanship (see 3.8).....	Visual	01005

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4.3.2.2 Plug (see dwg. 7549135 covering a detail of dwg. 7549133).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 1.00 percent		
101.	Outside diameter, min.....	Gage	02002
102.	Diameter of sleeve surface, min.	Gage	02003
103.	Length of leads, min.....	Gage	02004
104.	Length of lead insulation, max..	Gage	02005
105.	Plug cracked or damaged.....	Visual	02006
106.	Lead wire broken or not tinned..	Visual	02007
107.	Bridge missing or damaged to extent that function may be impaired.....	Visual	02011
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	02010

4.3.2.3 Sleeve (see dwg. 7549137 covering a detail of dwg. 7549133).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Outside diameter, max.....	Gage	03001
202.	Length, max.....	Gage	03002
203.	Evidence of poor workmanship (see 3.8).....	Visual	03003

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4.3.2.4 Plug assembly (see dwg. 7549134 covering a detail of dwg. 7549133).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 percent		
201.	Length, max.....	Gage	04001
202.	Sleeve loose.....	Manual	04002
203.	Evidence of poor workmanship (see 3.8).....	Visual	04003

4.3.2.5 Plug loading assembly, prior to applying priming charge (see dwg. 7549139 covering a detail of dwg. 7549133).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Flash charge missing.....	Visual	05001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	05002

4.3.2.6 Plug loading assembly (see dwg. 7549139 covering a detail of dwg. 7549133).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Explosive on exterior.....	Visual	06001
Major:	AQL 0.40 percent		
101.	Priming charge missing.....	Visual	06002

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	06003

4.3.2.7 Assembly (see dwg. 7549133).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup split or cracked.....	Visual	07001
2.	Lead wires not shorted.....	Visual	07002
3.	Insulation not removed on shorted lead wire.....	Visual	07003
Major:	AQL 0.65 percent		
101.	Total length.....	Gage	07004
102.	Diameter at plug end, max.....	Gage	07005
103.	Lead wire broken.....	Visual	07006
104.	Crimp not full 360 degrees.....	Visual	07007
Minor:	AQL 1.50 percent		
201.	Diameter of cup base, max.....	Gage	07008
202.	Plug cracked.....	Visual	07009
203.	Evidence of poor workmanship other than specified in Critical above (see 3.8).....	Visual	07010

4.3.2.8 Carton, prior to sealing (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Incorrect number of detonators..	Visual	08001
102.	Detonators in bundle not completely encased in foil.....	Visual	08002
103.	Pocket assembly torn or insecurely closed (when applicable).....	Visual	08003

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8).....	Visual	08004

4.3.2.9 Carton, sealed (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Carton damaged to extent that contents are exposed or liable to become exposed.....	Visual	09001
Minor:	AQL 0.65 percent		
201.	Marking missing, incorrect or illegible.....	Visual	09002

4.3.3 Production testing.- Unless otherwise specified herein, the sampling plans for the following tests shall be in accordance with MIL-STD-105.

4.3.3.1 Moisture content (see 3.3) - Major defect

TABLE I

Explosives	Code No.
Lead azide	10001
PETN	10002

The contractor shall provide adequate controls to insure that the explosives comply with the requirement. For verification, the contractor shall select and subject to test, a sample from that quantity of each explosive used in each eight hours of loading production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, the explosive represented by the sample shall be rejected. If components

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have been loaded with explosive containing excessive moisture, the remaining unloaded explosive and the loaded components shall be rejected. The test procedures shall be as specified in 4.4.1.1 and 4.4.1.2.

4.3.3.2 Functioning and waterproofness

4.3.3.2.1 Sampling.- One hundred and eighty-five (185) detonator assemblies shall be selected from each lot for test. The sequence of testing and number of samples to be subjected to each test is specified in Table II.

TABLE II

Sequence	Test	Classification	Code No.	No. of Samples	Test Paragraph
1.	Time (see 3.4.1)	Major	07011	35	4.4.2.1
2.	Waterproofness and Output (see 3.4.2, 3.5)	Major	07012	50	4.4.2.2
3.	Duds (see 3.4.3)	Major	07013	*100 less time and output duds	4.4.2.3

*Duds that occur in the time or output phase of testing shall be recorded and replaced from this sample.

4.3.3.2.2 Rejection.- The lot shall be rejected if any of the following test results occur:

a. Time.- The mean functioning time ($\bar{X}$) plus the product of 2.27 and the standard deviation (s) (see 6.5) exceeds the requirement specified in 3.4.1.

b. Waterproofness and Output.- One or more detonator assemblies fail to produce a perforation of the required size in the lead disc.

c. Duds.- The total number of detonator assemblies that fail to fire on the first attempt (dud) in the time, output and dud tests, equals or exceeds two (2).

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4.3.3.3 Resistance between shorted lead wires and cup (see 3.6) - Major defect - Code No. 07014.- This test shall be performed 100 percent in accordance with the procedure specified in 4.4.3. Any detonator assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.4 Resistance of lead circuit (see dwg. 7549133) - Major defect - Code No. 07015.- This test shall be performed 100 percent in accordance with the procedure specified in 4.4.4. Any detonator assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.3.3.5 Pull test of lead wires (see 3.7) - Major defect - Code No. 02001.- The sampling plan for this test shall be in accordance with Code Letter K and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.5 using equipment specified in 4.3.4.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 7549133 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Lead azide.- The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two gram (gr.) sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

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4.4.1.2 Moisture content of PETN.- The moisture content of the PETN shall be determined as specified in 4.4.1.1 except that the drying time shall be for one hour.

4.4.2 Functioning.- The detonator assemblies used for the time and dud tests shall be temperature conditioned to seventy (70) plus or minus ten (10) degrees Fahrenheit (F.) for two (2) hours minimum, prior to testing. Detonator assemblies subjected to the waterproofness and output test shall be conditioned for forty-eight (48) hours minimum at a two (2) to three (3) inch depth in water maintained at the above temperature.

4.4.2.1 Time.- The detonator assembly shall be assembled in the test fixture specified in 4.3.4 and fired by applying the energy from a .0022 microfarad capacitor, max. charged to three hundred (300) volts, max. The elapsed time between the application of the energy and the explosion shall be measured to determine compliance with the requirement.

4.4.2.2 Waterproofness and output.- The conditioned detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the same energy specified in 4.4.2.1. The diameter of the perforation in the .1345 inch, min., thick lead disc shall be measured to determine compliance with the requirement.

4.4.2.3 Duds.- The detonator assembly shall be assembled in the test fixture specified in 4.3.4 and fired by applying the energy specified in 4.4.2.1. Observation shall be made for duds.

4.4.3 Electrical resistance between shorted lead wires and cup.- The detonator assembly shall be assembled in the test fixture specified in 4.3.4 and the resistance measured when a five hundred (500) volt, sixty (60) cycle root mean square (r.m.s.) current is applied for two (2) seconds, max. Any assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.4.4 Resistance of lead circuit.- The detonator assembly shall be connected to the equipment specified in 4.3.4 and the lead circuit resistance measured when a current limited to twenty (20) microamperes is applied. Any assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

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4.4.5 Pull test of lead wires.- The plug shall be assembled in the equipment specified in 4.3.4 and the load applied. The wire and plug shall be observed for compliance with the requirement.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.

5.1.1.1 Rosette wrap.- A bundle of 25 detonators shall be made, lead wires out straight, using the lead wires from one detonator to secure the bundle. Each bundle of detonators shall be wrapped in a sheet of approximately 8 by 8 by .0015 inch thick aluminum foil complying with MIL-A-148, so that all detonators and lead wires in a bundle are completely encased by the foil. Five bundles of detonators, totaling 125 detonators, shall be packed in a folding carton, set-up box, or comparable container made of material with a minimum board thickness of .03 inch.

5.1.1.2 Pocket assembly, alternate pack.- The detonators shall be packed in pocket assemblies manufactured in accordance with Figure 1 (as applicable) made of .0015 inch aluminum foil complying with MIL-A-148. An adhesive seal shall be used to bond the faces of the aluminum foil together to form five pockets. The adhesive used for the sealing of the two faces of the aluminum foil shall be suitable for the application. The strength of the seal shall be sufficient to allow the insertion of the dowel (see Figure 1) without separation of the seal. The two faces of any seal shall not be misaligned by more than 1/16 inch when the pouch is formed. Each pocket shall be opened to form a cylinder using a smooth dowel with rounded end and similar to that shown on Figure 1. Each pocket may, however, be opened by other suitable methods which will result in providing a satisfactory pocket for insertion of the detonator without injury to the material. One M48 Electric Detonator shall be inserted, detonator end down, in each pocket. The lip of the pocket assembly shall be folded over to form a closed pouch. Twenty-five pocket assemblies or a total of 125 detonators shall be packed in a folding carton, set-up box, or comparable container made of a material with a minimum board thickness of .03 inch.

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5.1.1.3 Marking.- Each inner carton, set-up box or comparable container shall be marked: Patented - U. S. Patent 2,389,086.

5.2 Packing

5.2.1 Level C.- Unit packages shall be packed in accordance with the requirements of Code of Federal Regulations CFR 49 171-179 requirements for electric blasting caps and in a manner that will insure safe delivery from the manufacturer's plant to the first receiving activity.

5.2.2 Marking.- Marking shall be in accordance with dwg. 8796522. The DOT marking shall be: (QTY)* - ELECTRIC BLASTING CAPS - HANDLE CAREFULLY.

*Insert number of blasting caps in the exterior container.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an

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item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of ammunition data cards.- In addition to the normal distribution of records, one copy of all ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SMUPA-ND5
(2) SMUPA-DC.

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6.5 Check test for possible deterioration.- The following test is designed to prevent detonators that have deteriorated being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Two hundred (200) detonator assemblies shall be selected from each lot and tested as follows:

6.5.1 Functioning time and duds.- Forty (40) detonators shall be subjected to the test specified in 4.4.2.1. Observation shall be made for the functioning time and all duds that may occur shall be recorded. Duds will be replaced by detonators from the 200 selected. If the mean time ($\bar{X}$) plus 2.03 times the standard deviation (s) exceed 10 microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in 6.6 or its equivalent. Duds shall not enter into the calculation for standard deviation.

6.5.2 Waterproofness, output and duds.- Fifty (50) detonators shall be subjected to the test specified in 4.4.2.2. Observation shall be made of the perforation in the disc, and the duds that may occur shall be recorded. Duds shall be replaced by detonators from the 200 selected. If two or more detonators fail to adequately perforate the lead disc, the lot shall be rejected.

6.5.3 Duds.- The remainder of the detonator assemblies selected in 6.5 shall be subjected to the test specified in 4.4.2.3. The lot shall be rejected if the total number of duds accumulated in 6.5.1, 6.5.2 and 6.5.3 equals or exceeds four (4).

6.6 Calculation of sample standard deviation.- The following formula or equivalent shall be used in calculating the sample standard deviation:

$$s = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n-1}}$$

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where:

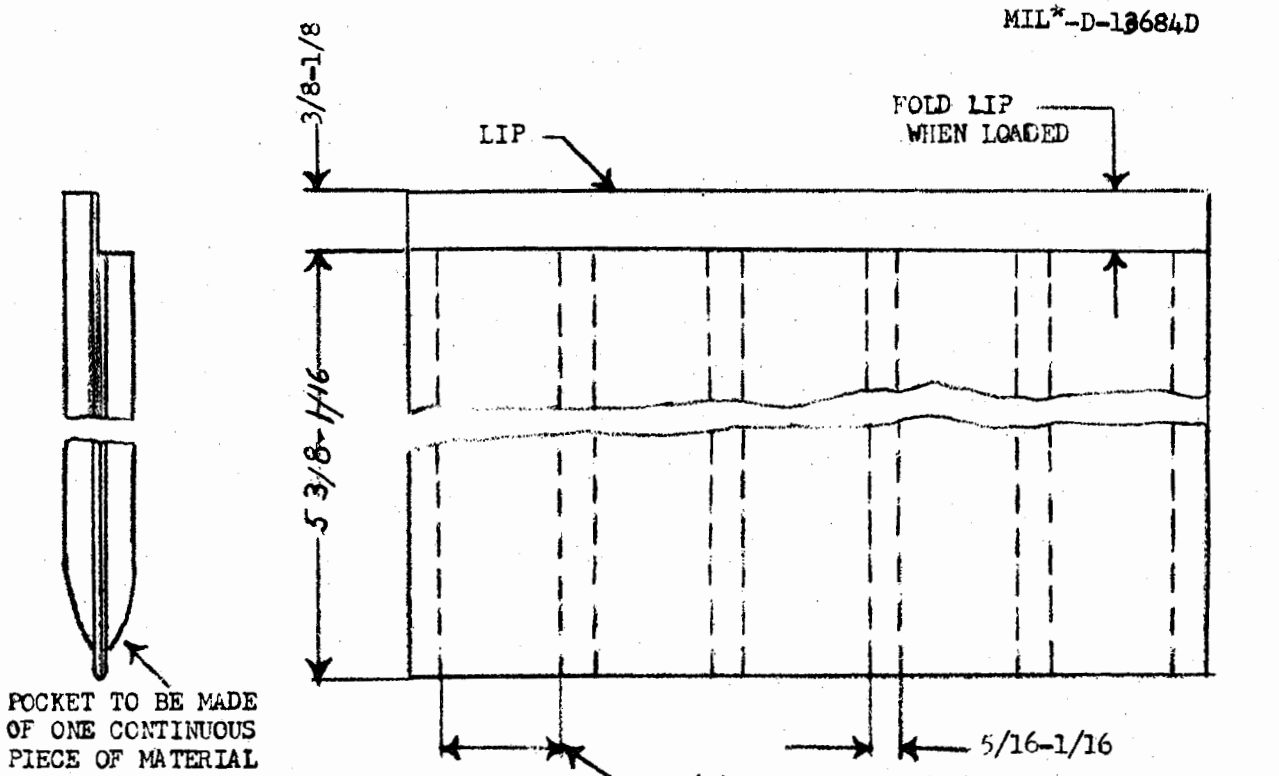
s = sample standard deviation.
x = numerical value of individual observations.
$\bar{x}$ = average value of the sample.
n = sample size.

Custodian:
Army- (AR)

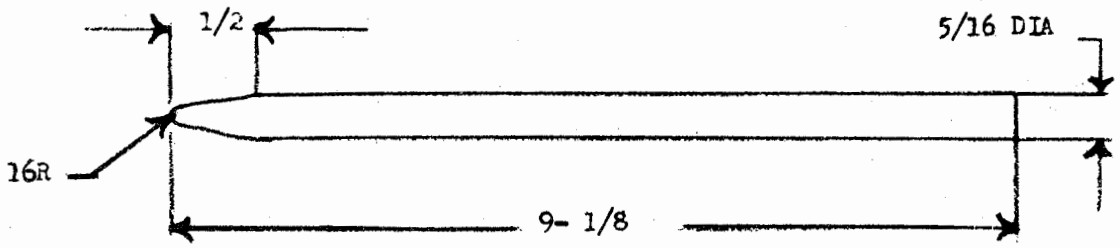
Preparing activity:
Army- (AR)

Project Number: 1375-A-020

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POCKET ASSEMBLY



DOWEL SHALL BE SMOOTH WITH NO SHARP CORNERS

DOWEL

FIGURE I

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